BIO 101 — Lecture 4: Photosynthesis

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Learning objectives

- Define photosynthesis and state its overall equation.
- Identify the two main stages: light-dependent reactions and the Calvin cycle.
- Explain the role of chlorophyll and the thylakoid membrane.
- Describe how ATP and NADPH link the two stages.
- Relate photosynthesis to cellular respiration.

1. What is photosynthesis?

- Process by which autotrophs convert light energy into chemical energy.
- Occurs in chloroplasts of plant cells.
- Overall equation: $6CO_2 + 6H_2O + \text{light} \rightarrow C_6H_{12}O_6 + 6O_2$.
- Carbon dioxide is reduced; water is oxidised.
- Two stages: light-dependent reactions and the Calvin cycle.
- Light reactions occur in the thylakoid membrane.
- Calvin cycle occurs in the stroma.

2. Chloroplast structure

- Double membrane envelope.
- Stroma: fluid-filled space where the Calvin cycle occurs.
- Thylakoids: flattened sacs stacked into grana.
- Thylakoid membrane contains photosystems I and II.
- Chlorophyll a and b are the main photosynthetic pigments.
- Accessory pigments: carotenoids, xanthophylls.
- Pigments absorb mainly red and blue light; green is reflected.

3. Light-dependent reactions

- Occur in the thylakoid membrane.
- Require light, water, ADP, NADP+.
- Photosystem II absorbs light, splits water (photolysis): $2H_2O \rightarrow 4H^+ + 4e^- + O_2$.
- Electrons travel down the electron transport chain.
- Proton gradient drives ATP synthase $\rightarrow$ ATP produced (photophosphorylation).
- Photosystem I re-energises electrons, reduces NADP+ to NADPH.
- Products: ATP, NADPH, O_2.

4. The Calvin cycle

- Occurs in the stroma; does not require light directly.
- Also called the light-independent reactions.
- Three phases: carbon fixation, reduction, regeneration.
- Carbon fixation: CO_2 + RuBP $\rightarrow$ 2 × GP (catalysed by rubisco).
- Reduction: GP + ATP + NADPH $\rightarrow$ G3P.
- Regeneration: G3P used to regenerate RuBP.
- For every 3 CO_2 fixed, 1 G3P is exported to make glucose.

5. Worked example

- Question: A plant fixes 18 molecules of CO_2. How many G3P are exported?
- Step 1: 3 CO_2 are needed per 1 G3P exported.
- Step 2: 18 ÷ 3 = 6.
- Answer: 6 G3P molecules are exported.
- Note: 5 of every 6 G3P are used to regenerate RuBP.

6. Limiting factors

- Light intensity — rate rises then plateaus.
- CO_2 concentration — rate rises then plateaus.
- Temperature — enzymes denature above optimum.
- Chlorophyll availability.
- Water availability (affects photolysis).
- Graph interpretation: plateau indicates another factor is limiting.

7. Comparison with respiration

- Photosynthesis: anabolic, endergonic, stores energy.
- Respiration: catabolic, exergonic, releases energy.
- Photosynthesis uses CO_2 and releases O_2.
- Respiration uses O_2 and releases CO_2.
- Both involve electron transport chains and ATP.
- Both use chemiosmosis to generate ATP.

8. Summary

- Photosynthesis converts light energy into chemical energy.
- Light reactions produce ATP and NADPH in the thylakoid membrane.
- The Calvin cycle uses ATP and NADPH to fix CO2 into G3P.
- Chlorophyll absorbs red and blue light.
- Limiting factors affect the rate of photosynthesis.